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**Section:** F

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### **Wiegers Requirement Prioritization**

**Scenario:** A University wants to develop a University Course Registration System that will allow students to manage their courses online. The system should simplify course selection, registration, and academic management.

**Stakeholder:** University Administration (Registrar Office)

Their goal is to manage course registration efficiently and avoid scheduling or enrollment issues.

#### **Requirements with Justification (According to the Stakeholder)**

| <b>Req</b> | <b>Requirement</b>                                     | <b>Relative Benefit</b> | <b>Relative Detriment</b> | <b>Stakeholder Reason</b>                                                                                            |
|------------|--------------------------------------------------------|-------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>R1</b>  | Students can <b>view available courses</b>             | 6                       | 3                         | Helpful for students but the university can still publish course lists through other channels (email/notice boards). |
| <b>R2</b>  | Students can <b>register for courses online</b>        | 9                       | 7                         | Core functionality. It automates registration and reduces manual workload for the administration.                    |
| <b>R3</b>  | Students can <b>drop or swap courses</b>               | 4                       | 7                         | Useful feature, but if missing it creates many manual requests and administrative burden.                            |
| <b>R4</b>  | Students can <b>view their timetable</b>               | 2                       | 1                         | Convenient feature but not critical since timetable could be shared through PDFs.                                    |
| <b>R5</b>  | Students receive <b>deadline notifications</b>         | 4                       | 9                         | Not implementing it can cause students to miss deadlines, creating complaints and administrative problems.           |
| <b>R6</b>  | Students can <b>view prerequisites and eligibility</b> | 4                       | 8                         | Important to prevent students from registering in incorrect courses which can create academic issues later.          |

**Step 1: Find Value total**

$$\text{Value Total} = \text{Relative Benefit} \times \text{Weight} + \text{Relative Determent} \times \text{Weight}$$

$$\text{Value Total} = \text{Benefit} \times \text{Relative Benefit Weight} + \text{Determent} \times \text{Relative Determent Weight}$$

**Step 2: Sum Value Total****Step 3: Find Value%**

$$\text{Value \%} = \frac{\text{Value total (Requirement)}}{\text{Sum value total}} \times 100$$

**Step 4: Find Cost%**

$$\text{Cost \%} = \frac{\text{Cost of Requirement}}{\text{Total Cost}} \times 100$$

**Step 5: Find Risk%**

$$\text{Risk \%} = \frac{\text{Relative Risk of Requirement}}{\text{Total Relative Risk}} \times 100$$

**Step 6: Find Priority Score**

$$\text{Priority} = \frac{(\text{Value\%})}{\text{Cost\%} \times \text{Relative Cost Weight} + \text{Risk\%} \times \text{Relative Risk Weight}}$$

$$\text{Priority} = \frac{\text{Benefit} \times \text{Relative Benefit Weight} + \text{Determent} \times \text{Relative Determent Weight}}{\text{Cost\%} \times \text{Relative Cost Weight} + \text{Risk\%} \times \text{Relative Risk Weight}}$$

If all weights = 1 then

$$\text{Priority} = \frac{\text{Benefit} + \text{Determent}}{\text{Cost} + \text{Risk}} = \frac{B + D}{C + R}$$

**Step 7: Rank**

Based on the priority, rank the requirements where 1 denotes the highest priority

| Weights      | 2                | 0.5                |             |        | 1             |       | 1             |       |          |      |
|--------------|------------------|--------------------|-------------|--------|---------------|-------|---------------|-------|----------|------|
| Requirements | Relative Benefit | Relative Detriment | Value Total | Value% | Relative Cost | Cost% | Relative Risk | Risk% | Priority | Rank |
| R1           | 6                | 3                  | 13.5        | 17.88  | 2             | 11.11 | 1             | 7.69  | 0.95     | 1    |
| R2           | 9                | 7                  | 21.5        | 28.48  | 4             | 22.22 | 2             | 15.38 | 0.76     | 2    |
| R3           | 4                | 7                  | 11.5        | 15.23  | 4             | 22.22 | 2             | 15.38 | 0.41     | 4    |
| R4           | 2                | 1                  | 4.5         | 5.96   | 1             | 5.56  | 1             | 7.69  | 0.45     | 3    |
| R5           | 4                | 9                  | 12.5        | 16.56  | 4             | 22.22 | 4             | 30.77 | 0.31     | 6    |
| R6           | 4                | 8                  | 12          | 15.89  | 3             | 16.67 | 3             | 23.08 | 0.40     | 5    |
| Total        | 29               | 35                 | 75.5        | 100    | 18            | 100   | 13            | 100   | 3.28     | 21   |